

Sarthak Samal

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EDUCATION

University of Colorado Boulder, Boulder, Colorado

B.S. in Biomedical Engineering

Minor in Electrical Engineering

Graduation Date: December 2024

Major GPA: 3.2

SKILLS

- SPICE simulation, MATLAB, C/C++, HTML, CSS, Python, LaTeX, JavaScript, SimVascular
- PCB Design, High-Speed Circuit Design, Prototyping Electronics, Signal Processing, Digital/Analog Electronics, CAD (SolidWorks certified), 3D Printing (MarkForge, Lulzbot, SLA), Arduino, Biomaterials, Project Management, Budgeting
- Microsoft Office Suite (including Excel VBA)
- English (Fluent), Odia (Fluent), Japanese (Intermediate)

ENGINEERING EXPERIENCE

Undergraduate Research Assistant | FLOWLab, University of Colorado Boulder

Spring 2023 – Present

- Used real patient DICOM CT angioscan data to construct individual 3D cardiovascular models from the aorta to the brain to determine the impact of distal aortic embolisms on strokes.
- Conducted CFD for vascular flow, concatenated DICOM scan slices in SimVascular, manually segmented major blood vessel paths.
- Simulated blood flow through meshed blood vessels in SimVascular, visualized .vti/.vtp files in software to determine velocity profiles of vascular flow.

Systems Engineer and Logistics Manager | Steadman Philippon, Boulder, Colorado

Fall 2023 – Spring 2024

- Collaborated on a senior capstone project to develop MATLAB image denoising machine learning algorithms. Enhanced the precision and effectiveness of bone joint X-ray image analysis pipelines, streamlining the process of generating 3D bone models by reducing computation time by 30%.
- Implemented a convolutional neural network-based approach to train an algorithm for X-ray image denoising and achieved an accuracy of 85% with our final model.
- Managed team operations, which included task scheduling, orchestrating regular check-in meetings, and facilitating seamless communication and collaboration among team members.

Medical Device Engineer | University of Colorado Boulder

Fall 2020

- Modeled, prototyped, and constructed Arduino-powered eyewear to provide real-time alerts to visually impaired users regarding nearby objects and people. Designed ultrasonic sensor arrangement to optimize field of view, connected to vibrational motors to alert the user of approaching obstacles/objects.
- Piped Arduino ultrasonic sensor data into MATLAB for more powerful signal processing, to develop more robust embedded algorithms for object detection.
- Conducted comprehensive market research to better understand the unique requirements of individuals with visual impairments, enabling the creation of an innovative product tailored to their specific needs.

LEADERSHIP & INVOLVEMENT

CU Boulder Milana, Boulder, CO | *Dancer*

Fall 2022 – Spring 2024

- Engaged in cultural dance ensembles committed to promoting the rich traditions of both classical and contemporary Indian dance forms within the university and across the state of Colorado. Showcased performances at diverse venues, including cultural festivals, exhibitions, and sporting events.

Colorado Crew Rowing Team, Boulder, CO | *Fundraiser, Varsity Rower*

Fall 2022 – Spring 2023

- Achieved Gold Medal in Varsity Men's 4+ Race
- Exemplified exceptional time management skills by consistently attending early morning practice sessions
- Collaborated with organizations and university facilities to raise \$1500+ in funds for the team.